

GENETICS 101 · WEEK 1 QUIZ

Genetics 101 — Week 1

Name: _____

Date: _____

How to play: Read each question twice, read every choice, cross out the wrong ones, then put an X in the box next to your answer. Get **6 of 8** right to earn the badge.

1. Where is a cell's DNA instruction book kept?

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- A. In the plasma of the blood, between the cells.
- B. In the nucleus of the cell.
- C. In the membrane around the cell.
- D. In the marrow inside the bones only.

2. DNA's code is written using how many different letters?

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- A. 4 (A, T, C, and G)
- B. 26 (A through Z)
- C. 2 (0 and 1)
- D. 20 (one per protein)

3. What is a gene?

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- A. One recipe in the instruction book — the instructions for building one protein.
- B. One of the 46 volumes that the whole instruction book is divided into.
- C. One single letter of the four-letter DNA code.
- D. A special cell that carries the instruction book from parent to baby.

4. How many chromosomes does a typical human body cell have, and where do they come from?

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- A. 92 — 46 from the mother and 46 from the father.
- B. 23 — all of them inherited from the mother alone.
- C. 46 — 23 from the mother and 23 from the father.
- D. 46 — all of them inherited from the father alone.

5. Identical twins happen when...

- ☐ A. a baby's cells divide much faster than usual during the first weeks in the womb.
- ☐ B. two babies are born on the same day in the same hospital to different mothers.
- ☐ C. two separate eggs are fertilized at the same time by two different sperm.
- ☐ D. one fertilized egg splits into two, so both babies share the same instruction book.

6. Why do twins so often end up in the NICU?

- ☐ A. Twins have to be kept together in one isolette for their first weeks.
- ☐ B. Twins are usually born early – about 6 in 10 arrive preterm – and often small.
- ☐ C. Twins are much more likely to catch jaundice from each other.
- ☐ D. Hospitals are required by law to observe all twins for a full week.

7. What is hemoglobin's connection to genetics?

- ☐ A. Hemoglobin is made by chromosomes floating in the blood.
- ☐ B. Hemoglobin is one of the four letters of the DNA code.
- ☐ C. Hemoglobin is a protein, built by following the instructions in a gene.
- ☐ D. Hemoglobin carries the DNA around the body inside red blood cells.

8. What was Rosalind Franklin's contribution to genetics?

- ☐ A. She invented the microscope powerful enough to see chromosomes.
- ☐ B. She proved that identical twins carry identical DNA.
- ☐ C. She discovered the first gene that controls a person's eye color.
- ☐ D. Her X-ray photograph of DNA revealed its twisted-ladder shape.

Score: _____ / 8

Badge earned? YES / NOT YET (need 6)